

DEXTER FERNANDES

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SUMMARY

Machine learning engineer with 3+ years deploying computer vision and deep learning models onto real-time, embedded platforms in production. Owned the full ML lifecycle, from data curation and training through quantisation, deployment and field validation, on industrial edge hardware including NVIDIA Jetson and Intel Movidius VPUs. MSc Robotics (Merit), with hands-on LiDAR SLAM and multi-sensor fusion on a mobile robot platform, alongside production experience processing multi-channel sensor telemetry. Works across computer vision, MLOps and embedded AI to move models from experimentation to reliable real-world operation.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP drilling rigs, from data curation and training through quantisation, deployment and field validation, improving detector F1 by 15% under 24/7 operation.
- Adapted YOLOv5 detectors in PyTorch for rig-specific classes, running Optuna sweeps and Weights & Biases tracking, and benchmarked against EfficientDet on a repeatable harness to catch regressions before release.
- Developed and maintained NVIDIA DeepStream 6.x and FFmpeg video pipelines for 24/7 embedded perception on BP drilling rigs, covering camera ingestion, inference and field diagnostics.
- Built ingestion and validation pipelines across image, video and sensor data with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout from 72h to under 24h.
- Built an activity classification workstream over 91 sensor channels using MultiRocket, gradient boosting and random forests for multivariate sensor telemetry classification.
- Led on-site validation on BP rig deployments, diagnosing model failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time embedded constraints.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Ran adversarial-condition training and validation simulating occlusion, lighting variation and motion artefacts, cutting performance variance by 25%.
- Used confusion-matrix failure clustering and TIDE error analysis on COCO-format production data to diagnose recurring detector failures and prioritise dataset fixes across camera feeds.
- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Designed a geometry-first cross-camera (MTMC) identity layer over per-camera DeepStream trackers, lifting pooled IDF1 from 0.198 to 0.245 on WildTrack.
- Instrumented the pipeline with structured JSON-line logging and a per-sensor health monitor tracking liveness, FPS vs expected and time-since-last-detection, flagging stalled streams in real time.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer with robot_localization fusing wheel odometry and IMU.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for embedded and edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.

- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.

EDUCATION

University of Bristol — MSc Robotics, Merit09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (rclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline, validating sensor placement for coverage.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier under JetPack, working within the platform's compute and memory limits for on-robot operation.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering07/2016 – 05/2020 · NAGPUR, INDIA

Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
Computer Vision & DL	object detection, image classification, semantic and instance segmentation, object tracking, optical flow, convolutional neural networks (CNNs), YOLO (v5/v8/v10/v26), RT-DETR, EfficientDet, PyTorch, TensorFlow, Keras, OpenCV, NumPy
Embedded & Edge	embedded AI, TensorRT, OpenVINO, ONNX, quantisation-aware training, FP16/INT8, structured pruning, layer fusion, NVIDIA Jetson AGX Xavier, Intel Movidius Myriad VPU, latency and throughput profiling
Video & Sensor Pipelines	NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, multi-camera synchronisation, multi-target multi-camera (MTMC) tracking, ISP and camera tuning (RGB and IR)
Robotics & SLAM	ROS2 (Robot Operating System 2), rclcpp, tf2, rosbag2, Gazebo, RTAB-Map, LIO-SAM, Cartographer, LiDAR SLAM, ICP (Iterative Closest Point), PCL (Point Cloud Library), Open3D, robot_localization, EKF
Sensor Data & Telemetry	LiDAR (Velodyne VLP-16), IMU, multivariate sensor telemetry classification, MultiRocket, gradient boosting, random forests
MLOps	MLflow, DVC (Data Version Control), Apache Airflow, Apache Kafka, Optuna, Weights & Biases, CVAT, synthetic data generation, benchmarking harnesses, drift tracking
Production & Cloud	Docker, CI/CD, GitHub Actions, Microsoft Azure, REST API design, containerised microservices, PostgreSQL, observability, Linux
Professional	technical leadership, mentoring, Agile and Scrum, stakeholder communication

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

Taco Bell — Service Champion / Team Trainer09/2024 – PRESENT · BRISTOL, ENGLAND

Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)

Best Speaker award, ABC Toastmasters Club (01/2020)